

CNC-X100 Vertical Machining Center Maintenance Manual

Product: CNC Machine | Model: CNC-X100 | Document Type: MAINTENANCE_MANUAL

1. Preventive Maintenance Philosophy & Schedule

Adhering to routine maintenance schedules prevents costly machine downtime, preserves positioning accuracy (-0.005 mm), and extends spindle lifespan.

2. Daily Inspection Tasks (Operator Duty)

- Coolant Tank: Inspect fluid level sight glass (maintain 80% fill level) and check concentration using optical refractometer (5% to 8%).
- Pneumatic FRL Unit: Check main regulator gauge reads 6.0 bar to 8.0 bar (nominal 6.5 bar). Drain condensate bowl.
- Automatic Lubrication Tank: Verify ISO VG 68 oil level above 50% capacity.
- Enclosure & Way Covers: Clean metal chips off X, Y, Z way covers and door glass using brass scraper and shop vacuum.
- Safety Systems: Test Emergency Stop button and safety door interlock operation.

3. Weekly Inspection Checklist (Maintenance Technician)

- Coolant Intake Filter: Remove screen basket from coolant tank and clean accumulated sludge with compressed air.
- Spindle Taper Inspection: Clean BT40 spindle internal taper using felt wiper tool. Check for fretting or galling.
- Tool Holder Inspection: Check retention knobs (pull studs) for cracking or wear (torque specification 45 Nm).
- Way Cover Wipers: Inspect rubber wipers on linear guide way covers for wear, tears, or chip embedding.

4. Monthly Maintenance Tasks

- Air Filter Replacement: Clean electrical cabinet intake filter mats; replace if clogged (Part # X100-FLT-002).
- Axis Drive Belts: Check tension of X and Y axis timing belts using acoustic meter (nominal 115 Hz).
- Automatic Tool Changer (ATC): Grease tool gripper arm pivot pins and cam box using NLGI Grade 2 lithium grease.
- Pneumatic Line Dryer: Replace 5-micron coalescing air filter element.

5. Cabinet Air Filter Replacement Procedure

Step-by-step instructions for electrical cabinet air filter maintenance:

1. Turn OFF main electrical disconnect breaker and open louver door covers on rear cabinet.
2. Remove filter retainer clips and pull out dusty filter mat.
3. Wash mat in mild detergent water, dry completely with compressed air hose, or insert new filter mat (Part # X100-FLT-002).

6. Coolant Inspection & Complete Flush Procedure

Every 6 months or 1,500 operating hours, perform complete coolant flush:

1. Pump out old coolant fluid into disposal tank.
2. Scrape metal chips and bio-sludge from 200L tank floor.
3. Flush lines with 2% system cleaner solution for 1 hour, drain completely, and refill with fresh 6% synthetic coolant emulsion to 80% fill mark.

7. Maintenance Interval Summary Matrix

Task Description	Frequency	Consumable / Part #
Check Lube & Coolant Leve	Daily	ISO VG 68 / Coolant
Clean Spindle Taper & Pul	Weekly	Felt Wiper Tool
Cabinet Air Filter Mat Sw	Monthly	Part # X100-FLT-002
ATC Cam Box Greasing	Every 3 Months	NLGI Grade 2 Lithium
Coolant Tank Complete Flu	Every 6 Months	System Flush Cleaner
Laser Interferometer Cali	Annual	Service Technician